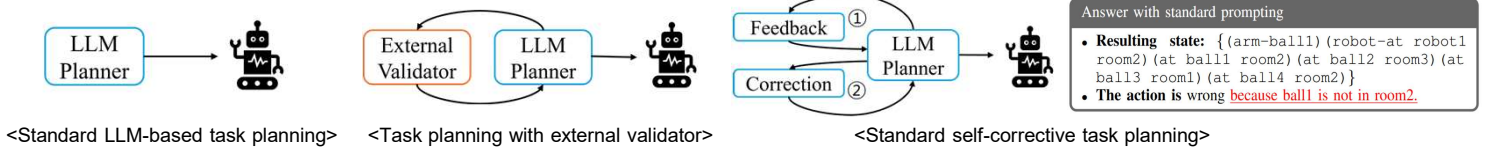


Self-Corrective Task Planning by Inverse Prompting with Large Language Models

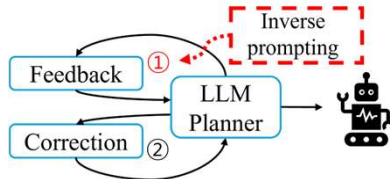
Introduction & Motivation

- Standard LLM-based task planning**
 - The LLM planner generates action sequences based on a given situation, but it does not validate them
- Task planning with external validator**
 - The LLM planner generates action sequences and validates them with additional efforts
 - Construct a predefined set of error types and the corresponding feedbacks
 - Leverage external knowledge resources for retrieval
- Standard self-corrective task planning**
 - The LLM planner first generates action sequences, then validates and refines them by providing feedback in order to correct them on its own
 - Standard prompting
 - Generate the feedback with a single reasoning step
 - Lack thorough grounding or steps that support a clear reasoning process

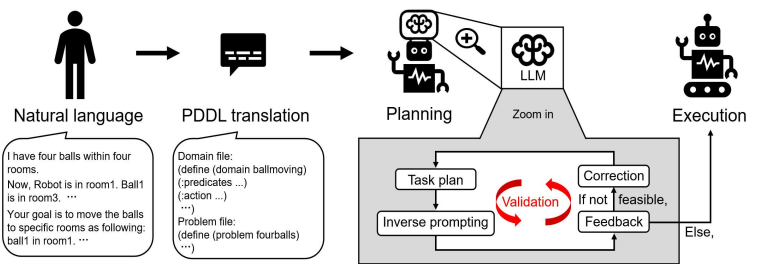


Proposed Method: InversePrompt

- Multi-step self-corrective reasoning process**
 - Explicitly validate the logical flow of the generated plan



- Proposed framework**



- Example of InversePrompt**

Question

- Current state:** {(robot-at robot1 room1) (at ball1 room2) (at ball2 room3) (at ball3 room1) (at ball4 room2)}
- Goal state:** {(at ball1 room1) (at ball2 room1) (at ball3 room3) (at ball4 room4)}
- Examined action:** (pick ball1 room2)

Answer with inverse prompting

- Resulting state:** {(arm-ball1) (robot-at robot1 room2) (at ball1 room2) (at ball2 room3) (at ball3 room1) (at ball4 room2)}
- Inverse action:** (drop ball1 room2)
- Inverted state:** {(robot-at robot1 room2) (at ball1 room2) (at ball2 room3) (at ball3 room1) (at ball4 room2)}
- Difference:** (robot-at robot1 room2) & (robot-at robot1 room1)
- The action is wrong** because robot1 is in room1, not room2.

Experimental Results

- Benchmark experiments**

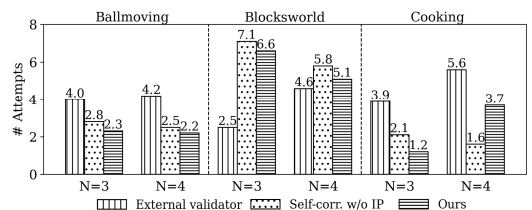
- Comparison of success rates

Scenario		GPT-4o-mini		
		No Validator	External Validator	Self-corr. w/o IP
Ballmoving	N=3	70%	85%	85%
	N=4	50%	75%	100%
Blocksworld	N=3	75%	90%	75%
	N=4	55%	60%	50%
Cooking	N=3	45%	65%	50%
	N=4	30%	45%	30%
Average		54%	70%	61%

- Performance of self-correction

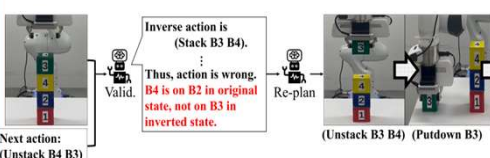
Scenario		GPT-4o-mini		Gemini-1.5-Flash	
		Self-corr. w/o IP	Ours	Self-corr. w/o IP	Ours
Ballmoving	N=3	80%	100%	70%	85%
	N=4	80%	100%	25%	55%
Blocksworld	N=3	30%	35%	90%	100%
	N=4	60%	60%	85%	100%
Cooking	N=3	55%	70%	80%	85%
	N=4	30%	35%	85%	100%
Average		56%	67%	73%	88%

- The number of attempts for correction



- Real-world experiments**

- Blocksworld



- Cooking

